

Mehrnaz Ayazi

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OBJECTIVE

Researcher and engineer specializing in **computer graphics** and **physics-based simulation**, driven by impactful research that **bridges theory and practice**. Dedicated to **transparent, supportive, and equity-minded teaching** that empowers students to **master complex material**. Seeking roles that combine research excellence with **rigorous** teaching.

EDUCATION

PhD, Computer Science, University of California, Riverside (UCR) Jan 2026

M.S. in Computer Science, University of California, Riverside (UCR) Jan 2023

Bachelor of Science in Computer Engineering, Amirkabir University of Technology (AUT) Jul 2019

TECHNICAL SKILLS

Languages: C/C++, Python, HTML/CSS, SQL, Java, VHDL, Verilog, MATLAB

Natural Languages: Farsi(Native), French(Working proficiency) , Arabic (familiar)

Tools: Shell Scripting, MongoDB, MySQL, Linux (Ubuntu), Houdini, Tensorflow, Pytorch, Django, Docker

PROJECTS AND PUBLICATIONS

Simplified conservative discretization of the Cahn-Hilliard-Navier-Stokes equations,

Jason Goulding, **Mehrnaz Ayazi**, Tamar Shinar, Craig Schroeder,

Journal of Computational Physics, Volume 519, 2024, 113382, ISSN 0021-9991,

<https://doi.org/10.1016/j.jcp.2024.113382>.

- The paper introduces a novel, second-order accurate, mass-momentum consistent discretization for the Cahn-Hilliard Navier-Stokes equations, featuring a simplified and adjustable surface tension model for sharp interface simulations.
- Debugged code, ensured mathematical accuracy, and developed solutions with validated testing for reliability. (PhysBAM, C++)

Buoyancy-driven Phase Separation in the Material Point Method

Mehrnaz Ayazi, Craig Schroeder, Tamar Shinar

- Introduced phase separation to the Material Point Method (MPM) in PhysBAM in C++ by introducing a multigrid scheme, addressing the challenge of simulating multi-phase fluids in MPM.
- Introduce a phase separating potential energy, resulting in phase separation in the absence of gravity
- Introduced a novel multi-phase level set projection scheme to render two-phase flows.
- This addition broadens MPM's capabilities in handling materials that exhibit both solid and fluid characteristics.

Phase Separation in the Cahn-Hilliard Equation in the Material Point Method

- Introduced phase separation to the Material Point Method (MPM) in PhysBAM in C++ by incorporating the Cahn-Hilliard equation, addressing the challenge of simulating two-phase fluids.
- This addition broadens MPM's capabilities in handling materials that exhibit both solid and fluid characteristics.

Multi-style Lagrangian Neural Style Transfer

- Developed a method to style fluid simulations using multiple style images based on selected attributes (e.g., velocity, turbulence, artist guides).
- Aimed to enhance artist control, temporal coherence, and precision in localized stylization. (Python, Tensorflow)

Formal Specification and Implementation of an Electronic Payment System(Undergraduate Thesis)

- Designed, specified, and implemented an EPS using Z specification language and Python programming language. (Z, Python, HTML, CSS, Django)

EXPERIENCE

TEACHING

Department of Computer Science, UCR

Teaching Assistant/Discussion Section Leader/Lab Instructor

- [Scientific Computing](#)(CS210) (Grad: Fall 2022, Fall 2023)
- [Computer Graphics](#)(CS230) (Grad: Winter 2023)
- [Introduction to Computing](#) (CS008)(Winter 2022)
- [Principles of Web Development](#) (CS110)(Spring 2022)
- [Computer Graphics](#) (CS130)(Summer 2022)
- [Data Analysis Methods](#) (CS105)(Fall 2024, Winter 2025)
- [Discrete Structures](#) (CS111) (Spring 2025) (**Head TA**)

Lecturer on record

Department of Computer Science, UCR

- [Data-Oriented Introduction to Computing 1](#)(CS009A) (Summer 2025)
- [Discrete structures](#)(CS111) (Summer 2025)

Responsibilities

- Graded assignments and exams, led lab sessions, and held office hours to support student learning.
- Taught subjects including Linear Algebra, Numerical Analysis, C++, Web Development, Computer Graphics, and Data Science.

Productions Software R&D Team, DreamWorks Animation,

Summer 2023; Summer 2024

Production Software Intern

Burbank, CA

- Conducted an extensive literature review to assess contemporary methodologies in hair simulation, identifying key trends and challenges, which led to a refined understanding of current limitations and potential areas for improvement in simulation techniques.
- **Developed a prototype for real-time hair simulation by implementing a novel method from ACM SIGGRAPH 2023**, leveraging GPU acceleration to achieve interactive performance. The prototype was then used to evaluate the method's feasibility for integration into an actual production pipeline, providing insights into its practical application and potential for enhancing production efficiency.
- Research & Development Intern (**Summer 2024**), advancing the Multi-style Lagrangian Neural Style Transfer project.
- Collaborated with FX artists to develop a neural style transfer tool for fluid simulations in Houdini, enhancing usability and creative control.

Max Planck Institute for Software Systems (MPI-SWS)

Research Intern

March 2020 -Jun 2020

- Collaborated on a three-month research project in **reinforcement learning** and **machine learning**.
- Focused on developing and applying methods to ensure that **agents** in a **distributed system provide truthful and accurate data** from their surroundings.
- Achieved the targeted results by leveraging a fairness-based approach using **Shapley values**, ensuring equitable contribution assessments. The work, conducted under the guidance of Dr. Goran Radanovic,
- Improved reliability in data aggregation and decision-making within the system.

Shanghai University of Finance and Economics (SUFE)

Visiting Scholar

Dec 2019

- Engaged in research focusing on **stochastic learning challenges** caused by uncertainty and incomplete data
- during a one-month visiting scholar program at SUFE, with the valuable guidance and supervision of **Dr. Nick Gravin**.

CafeBazaar Summer Camp

Intern

Jun - Aug 2018

- Collaborated on a website for programming classes targeting 30M Persian-speaking users; built a Django-based coding environment to compile and evaluate code with instructor test cases.

LEADERSHIP

President, Graduate Student Association of Computer Science Department, UCR (2023-2024)

- Advocated for graduate students in the Computer Science and Computer Engineering Department at UCR, serving as a liaison between students, faculty, and staff.
- Organized quarterly and weekly events to improve graduate students' morale, networking skills, and opportunities for engagement with faculty, staff, and peers.
- Developed and led events for graduate students to present their research, preparing them for defense, qualifying exams, and conference presentations.
- Fostered a supportive academic community by facilitating connections and open communication among students, faculty, and staff.

Mentor, UCR's International Student Peer Mentor Program (ISPMP) (2023-2024)

- Assisted first-year international students with their integration into life in the U.S. during the 2023-2024 academic year.
- Assigned two mentees, holding bi-weekly meetings to review and support their progress toward set goals.
- Guided accessing resources and answered questions related to academic and personal life aspects in the U.S.

Volunteer

- Volunteer at MESA UCR (various quarters)
- IT Systems Admin at CEIT Computer Site at Amirkabir University of Technology (Jan - Jul 2016)
- Technical Staff in the Linux Festival held by the Computer Engineering Department, Amirkabir University of Technology

AWARDS AND CERTIFICATES

- University of California, Riverside, University Teaching Certificate
- Awarded a full scholarship by the UC Riverside Computer Science Department for the Grace Hopper Conference (2023)
- Awarded a full scholarship by the Computing Research Association to attend CRA-WP Cohort 2023 and 2025
- UCR Gluck fellow

RELEVANT COURSES

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| • Artificial Intelligence | • Game Theory |
| • Data Mining techniques | • Graph Theory |
| • Scientific Computing | • Statistics |
| • Design and Analysis of Algorithms | • Engineering Economics |
| • Numerical Analysis | • Convex Optimization |
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